



Fault inference diagnosis based on causal network and large language model-enhanced knowledge graph network

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ABSTRACT

Current research in mechanical fault inference diagnosis primarily relies on knowledge graph networks (KGN) for reasoning. Compared to traditional diagnostic methods, KGN offers advantages in knowledge representation, but it suffers from insufficient causal mining and limited reasoning capability. To this end, a fault inference diagnosis method (CN-LLM-KGN) that integrates causal networks (CN), large language models (LLMs), and KGN is proposed. Firstly, the causal structure and key causal paths of fault propagation are mined from the monitoring data through CN. Domain knowledge constraints are introduced to correct and optimize the traditional Peter-Clark algorithm. Secondly, the LLMs are utilized to enhance the KGN, including entity/relation completion of the knowledge graph, semantic alignment, and optimization of reasoning rules, to address the problems of knowledge sparsity and reasoning rigidity in traditional knowledge graphs. Finally, the constructed causal inference module, via an attention-weighted fusion strategy, drives the LLM to achieve causal traceability inference, knowledge-based fault diagnosis, and precise root cause localization (RCL). The experiment showed that the RCL accuracy of the CN-LLM-KGN method reached 81.12 %, providing a new path for the intelligent diagnosis of rotating machinery systems. This study proposes a novel intelligent diagnostic architecture that possesses strong causal reasoning capabilities, enabling effective fault traceability and precise localization, thereby forming an efficient and reliable hybrid intelligent diagnostic system.

1. Introduction

Under the technological wave of the deep integration of Industry 4.0 and intelligent manufacturing, mechanical systems are rapidly evolving in a complex direction characterized by deep coupling of multiple components, dynamic and diverse working conditions, and massive data interaction [1,2]. From the multi-axis coordinated transmission of the gearboxes in wind turbines to the dynamic load coordination of the traction systems in high-speed trains, the structural complexity and operational mechanism complexity of the equipment have both shown an exponential growth trend [3,4]. The stable operation of such critical mechanical systems directly determines the efficiency of industrial production, the control effect of operation and maintenance costs, as well as the safety level of the working environment [5,6]. Therefore, the construction of a precise, efficient, and traceable fault diagnosis system has become a core technical requirement for ensuring the reliable operation of intelligent manufacturing systems.

The core objective of mechanical fault diagnosis is not only to achieve early detection and warning of abnormal equipment conditions, but more importantly, to accurately identify the propagation path of the fault and locate the root cause of the fault. This goal places even higher demands on the causal analysis capabilities and knowledge support capabilities of the diagnostic methods. The traditional mechanical fault diagnosis methods mainly rely on a model that combines signal processing with expert experience [7–10]. Zhang et al. [11] proposed a new active fault diagnosis framework consisting of an auxiliary input generation component, a fault diagnosis observer, and a signal analysis unit. Lv et al. [12] proposed an improved extreme learning machine to achieve intelligent fault diagnosis for rolling bearings under various operating conditions. Bi et al. [13] proposed a unified diagnostic framework based on vibration source inversion, which is used to handle fault signals of transmission components. Although these methods can achieve basic fault identification in scenarios

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with simple working conditions and clear fault patterns, they have significant limitations in complex industrial environments. On the one hand, due to the lack of modeling of the causal relationship of faults, it is difficult to distinguish the causal correlations among data from false correlations, resulting in generally low accuracy rates for identifying the root causes of faults. On the other hand, the diagnostic results are highly dependent on the accumulated domain experience of experts. When dealing with new and unfamiliar faults, there is a high risk of misjudgment or missed detection.

In recent years, deep learning technology, with its ability of autonomous feature learning and deep feature extraction, has provided a new path to break through the limitations of traditional methods. A convolutional neural network (CNN) can automatically extract fault spatial features from time-frequency diagrams [14]. Recurrent neural networks (RNNs) and their variants can capture the temporal evolution patterns of faults [15]. The transformer model based on the attention mechanism can model the long-range dependencies of multi-sensor signals [16]. However, deep learning methods still encounter a thorny black box problem in practical industrial applications [17]. The model only outputs the fault classification results, but it is unable to explain why it was determined to be that particular fault. Deep learning requires a large amount of labeled data [18]. However, in industrial scenarios, the sample sizes of new or rare faults are often insufficient to obtain sufficient fault features [19]. More importantly, the essence of deep learning models is that they learn the patterns of data distribution, rather than the physical causal mechanisms of mechanical systems.

The groundbreaking achievements of large language models (LLMs) in fields such as natural language processing and computer vision have brought new technological opportunities for the upgrading of mechanical fault diagnosis technology [20,21]. Liao et al. [22] proposed a data augmentation method assisted by an LLM. Chen et al. [23] proposed a new cognitive diagnosis model based on the LLAMA3-70B framework. The powerful semantic understanding and knowledge integration capabilities of large models enable the integration of unstructured text and structured data, thereby constructing a cross-modal fault cognition system. The general knowledge accumulated during the pre-training stage can be adapted to the fault diagnosis task with only a small amount of domain-specific samples for fine-tuning, thereby alleviating the scarcity of samples in industrial scenarios [24]. However, when directly applying general large models to mechanical fault diagnosis, there are still problems, such as the difficulty in extracting causal features from the data and insufficient utilization of domain knowledge.

To address these challenges, this paper proposes a causal inference

fault diagnosis method (CN-LLM-KGN) based on causal networks (CN) and enhanced knowledge graph networks (KGN) with LLM. The core contributions of this article are:

- (1) A collaborative enhancement mechanism for CN and KGN was proposed to improve the accuracy and interpretability of fault diagnosis and root cause localization.
- (2) A causal inference loss function was constructed, and causal constraints were incorporated into the training process of the LLM to ensure that the diagnostic results conform to the causal mechanism of the physical system.
- (3) Utilizing LLM to enhance the representation and reasoning capabilities of knowledge graphs and verifying the effectiveness of the method in scenarios with few-shot samples.

The experimental results show that the proposed method significantly outperforms the existing baseline methods in the root cause localization (RCL) task, providing a new paradigm for intelligent diagnosis of rotating machinery systems.

The rest of the article is described as a fellow. The proposed CN-LLM-KGN causal inference fault diagnosis method is introduced in Section 2. Detailed experimental design is described in Section 3. The results analysis and discussion are described in Section 4. Section 5 is the conclusion.

2. CN-LLM-KGN causal inference fault diagnosis method

The overall framework of the CN-LLM-KGN method is shown in Fig. 1, which consists of three core modules: CN [25] construction and causal feature extraction, a KGN [26] enhanced by an LLM, and a causal reasoning fusion diagnosis module.

Step 1. Data preprocessing: Collect multi-source monitoring data of the mechanical system and extract the time-frequency domain features of the signals using continuous wavelet transform (CWT), providing input for subsequent causal discovery.

Step 2. Construction of CN: Based on the improved Peter-Clark (PC) algorithm, an initial causal directed acyclic graph (DAG) is learned from the preprocessed data, and domain knowledge is integrated to correct the DAG. Further, the causal effect network (CEN) is utilized to quantify the strength of each causal edge in the graph, and a weighted causal graph is constructed.

Step 3. Causal feature extraction: Identify the key causal paths during the fault propagation process, extract the causal strength of

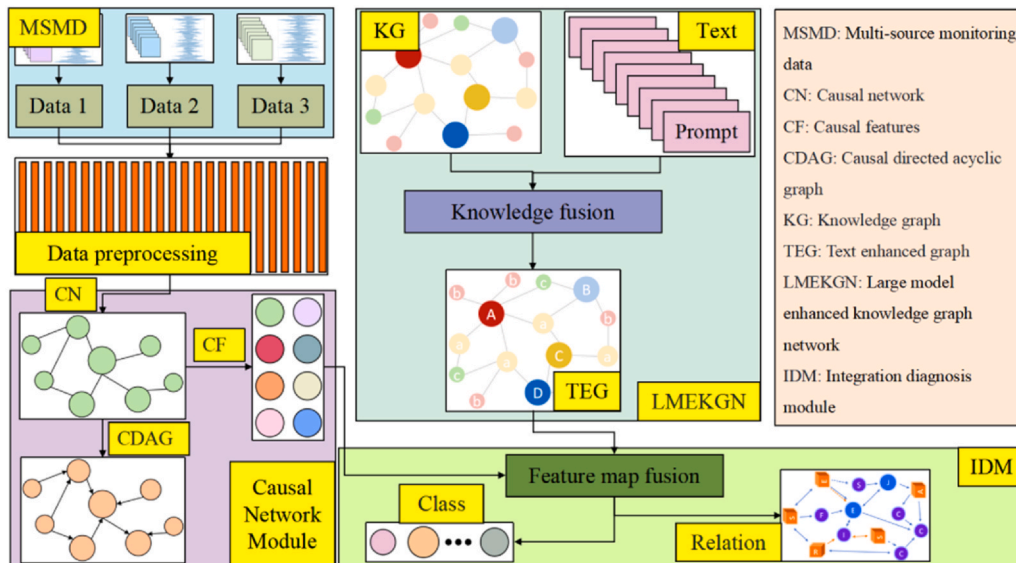


Fig. 1. The proposed CN-LLM-KGN for causal inference diagnosis.

variables along the paths to form the path feature vector, and calculate the average treatment effect (ATE) between this vector and various faults to construct the causal effect feature. Combine these two to form the final causal feature vector F_{CN} .

Step 4. Knowledge graph enhancement: Construct a mechanical fault knowledge graph centered on equipment-components-faults-parameters-repair, using the TransE algorithm to generate multi-dimensional entity relationship embeddings E_{KGN} . The fine-tuned LLM enhances the knowledge graph through a three-stage strategy, comprising knowledge completion, semantic alignment, and reasoning rule optimization, to obtain the refined embedding representation $E_{enhanced}$.

Step 5. Feature fusion: A fusion strategy based on the attention mechanism is designed, which adaptively weights and integrates the causal feature F_{CN} and the enhanced knowledge graph embedding $E_{enhanced}$ to form a fused feature F_{fusion} with multi-source semantics.

Step 6. Causal inference and diagnosis: The fused feature F_{fusion} is input into the fine-tuned large model, and its enhanced causal reasoning ability is utilized to complete fault diagnosis and root cause location.

2.1. Construction of causal networks and extraction of causal features

2.1.1. Data preprocessing

First, the collected vibration signals are preprocessed and segmented using a sliding window, as shown in Fig. 2. Here, k represents the length of the data, and n represents the sample size. It is particularly important to note that this study employs the interval independent sampling method. By collecting data samples that do not overlap within specific intervals, the inherent correlations between the data are completely eliminated, thereby effectively preventing the occurrence of interactions. For each vibration signal within the window, a CWT is performed using the complex Morlet wavelet [27] at multiple logarithmic scales. Extract its amplitude coefficient and obtain the low-dimensional feature vector through grouping mean dimensionality reduction. Ultimately, each sample is composed of multiple consecutive wavelet features, forming a feature sample set.

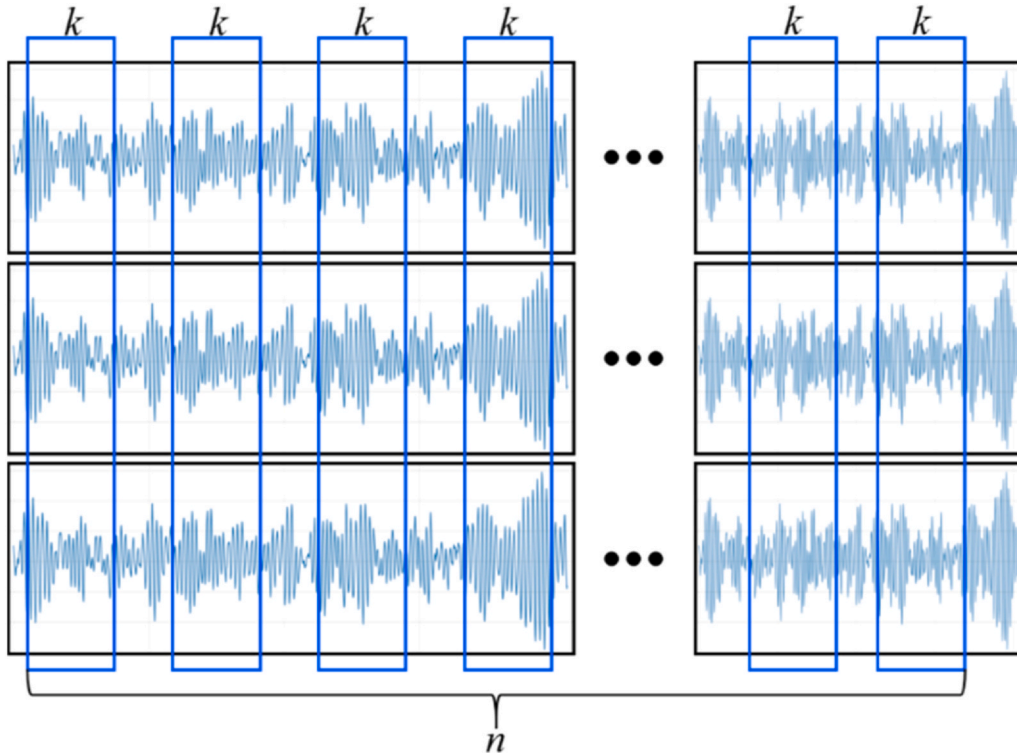


Fig. 2. Data preprocessing of signals.

2.1.2. Construction of causal networks

Traditional fault diagnosis methods often overlook the causal mechanism of fault propagation, resulting in diagnostic outcomes that lack interpretability and physical consistency. CN is capable of extracting causal relationships among variables from monitoring data, revealing the fault propagation paths, and providing a physically interpretable reasoning basis for diagnosis. This study employs the improved PC [28] algorithm and incorporates domain knowledge constraints to enhance the accuracy and rationality of the causal graph.

The improved PC algorithm is used to learn the initial DAG from the standardized preprocessed data. This method is based on the Fisher Z condition independence test. Its statistical decision criterion can be expressed as:

$$I(X, Y|Z) \Leftrightarrow \hat{p}(X, Y|Z) = \hat{p}(X|Z)\hat{p}(Y|Z), \quad (1)$$

where $I(X, Y|Z)$ denotes the conditional independence of X and Y given Z , and $\hat{p}$ represents the conditional probability distribution.

Based on the knowledge of the mechanical field, the initial DAG is corrected by eliminating the unreasonable edges, ensuring that the causal structure conforms to physical laws.

The strength of each causal edge is calculated using CEN. The causal effect (CE) value is calculated as follows:

$$CE(X \rightarrow Y) = \frac{1}{n} \sum_{i=1}^n [Y_i - Y_i(X_i = x_0)], \quad (2)$$

where n represents the sample size, and Y_i is the observed outcome, and $Y_i(X_i = x_0)$ represents the counterfactual outcome when X_i is set to the baseline value x_0 .

Based on the directed edge information, an adjacency matrix is constructed and normalized to form the final weighted causal graph, as shown in Fig. 3.

2.1.3. Causal feature extraction

Identify the core causal paths of fault propagation, and combine the causal strength values of all variables along the paths to form the path feature vectors. In the causal graph, there are m core causal paths

This study utilized Llama2-7B as the base large language model [31]. This model achieves a good balance between the number of parameters and inference efficiency. With 7B parameters, it is capable of understanding and generating complex fault diagnosis semantics, and its computational cost is within the feasible range for industrial applications. The fine-tuning of the model is carried out using the low-rank adapter (LoRA) technology, which is a parameter-efficient fine-tuning method. By introducing a LoRA to approximate the model weight updates, it enables general large models to adapt to professional domain tasks with extremely low training costs.

For a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA restricts its update process to be represented by a low-rank decomposition matrix BA :

$$h = W_0x + \Delta Wx = W_0x + BAx, \quad (10)$$

where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ are trainable adapter parameters, and the rank $r \ll \min(d, k)$. During the training process, the original weights W_0 are frozen, and only the adapter parameters A and B are updated.

The model is supervised fine-tuned using the standard cross-entropy loss function:

$$\mathcal{L}_{\text{LLM}} = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \log P(w_t^{(i)} | w_{<t}^{(i)}, \Theta_{\text{LoRA}}), \quad (11)$$

where Θ_{LoRA} represents all the LoRA adapter parameters, and $w_t^{(i)}$ is the target label of the t -th word in the i -th sequence.

By enhancing the representation of knowledge graphs through pre-trained language models, the aim is to address the semantic sparsity, limited relationship representation capabilities, and insufficient domain knowledge integration issues that exist in traditional knowledge graphs in the field of mechanical fault diagnosis. The enhanced knowledge graph can deeply integrate text semantics with structured knowledge, enhance the reasoning ability of entity relationships, provide a more accurate computable knowledge foundation for fault traceability and diagnostic decision-making, and thereby support subsequent deep reasoning and explainability analysis. The specific process is as follows:

- (1) Input the description text of the entities and relationships into the language model to obtain the semantic embeddings:

$$e_{\text{semantic}} = \text{LLM}(\text{description}). \quad (12)$$

Align the semantic embeddings with the original graph embeddings through the projection layer:

$$E_{\text{enhanced}} = \text{MLP}(e_{\text{semantic}}). \quad (13)$$

- (2) Select the entity with the highest similarity for completing the similarity relationship:

$$\text{score}(h, r, t) = \|h + r - t\|. \quad (14)$$

The total loss of the training process is expressed as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{TransE}} + \lambda \cdot \mathcal{L}_{\text{reg}}, \quad (15)$$

where $\mathcal{L}_{\text{reg}}$ represents the L2 regularization loss.

2.3. Causal inference integration diagnostic module

2.3.1. Feature fusion and classification

The causal feature F_{CN} and the knowledge graph feature E_{enhanced} are processed through two separate processing networks. Each network $\mathcal{H}(\cdot)$ consists of fully connected layers, batch normalization, and dropout layers, and uses L2 regularization to prevent overfitting. The expressions are as follows:

$$H_{\text{causal}} = \mathcal{H}_{\text{causal}}^{(2)}(F_{\text{CN}}), \quad (16)$$

$$H_{\text{kg}} = \mathcal{H}_{\text{kg}}^{(2)}(E_{\text{enhanced}}). \quad (17)$$

The processed features are calculated for their correlations through the attention function Γ and then a weighted fusion feature is generated:

$$F_{\text{fusion}} = \Gamma(H_{\text{causal}}, H_{\text{kg}}). \quad (18)$$

The fused features are input into the softmax network φ to calculate the probability distribution of the fault types:

$$P(\text{fault} | F_{\text{fusion}}) = \varphi(F_{\text{fusion}}). \quad (19)$$

2.3.2. Causal consistency constraints

The LLM has powerful semantic understanding and generation capabilities. It can extract fault semantic information from unstructured text and deeply integrate it with structured knowledge graphs. To achieve precise fault diagnosis and root cause localization, the research adopts an end-to-end joint training strategy. Integrate the classification task and the causal structure constraints into a unified optimization framework. The loss function of this method consists of two parts:

$$\mathcal{L}_{\text{uni}} = \mathcal{L}_{\text{cls}} + \eta \cdot \mathcal{L}_{\text{causal}}, \quad (20)$$

where $\mathcal{L}_{\text{cls}}$ represents the cross-entropy loss function for fault classification; $\mathcal{L}_{\text{causal}}$ represents the causal consistency constraint loss, and η is a hyperparameter used to balance the contributions of the two losses.

The classification loss $\mathcal{L}_{\text{cls}}$ is defined as:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \cdot \log(\hat{y}_{i,c}), \quad (21)$$

where N represents the sample size; C represents the fault category, and $y_{i,c}$ indicates the true label of sample i belonging to category c , while $\hat{y}_{i,c}$ represents the predicted category probability by the model.

The causal consistency loss $\mathcal{L}_{\text{causal}}$ achieves this by constraining the causal distribution predicted by the model to be consistent with the prior causal graph structure:

$$\mathcal{L}_{\text{causal}} = \frac{1}{N} \sum_{i=1}^N \|\hat{C}_i - C_i\|^2, \quad (22)$$

where $\hat{C}_i$ represents the causal strength matrix predicted by the model, while C_i is the adjacency matrix of the real causal graph constructed based on domain knowledge.

3. Experiments

As shown in Fig. 5, the core of the CWRU platform is an electric drive system, which is powered by an electric motor, measured by torque sensors and encoders, and subjected to simulated loads by a dynamometer. The objects being tested are the bearings at the drive end and the fan end, which are key components for the fault diagnosis research. The experiment utilized the publicly available CWRU bearing dataset, which includes normal conditions, inner ring wear, outer ring wear, and rolling element wear faults. The detailed information is shown in Table 1. The training samples for each category are 50, the test samples are 100, and the length of the samples is 2048. The size of the faulty samples is all 0.007 inch. The load is 2 horsepower, the rotational speed is 1750 r/min, and the sampling frequency at the driving end is 48 kHz. The signals are divided into time windows, and at each point of data, the window is regarded as a sample, with the fault type labeled accordingly. The detailed model information is shown in Table 2.

To analyze the causal inference ability of the model under different feature extraction conditions, the feature settings as shown in Fig. 6 were carried out. H represents the features obtained through continuous wavelet transformation [32], T represents the features obtained based on H , and Z represents the time-domain or frequency-domain

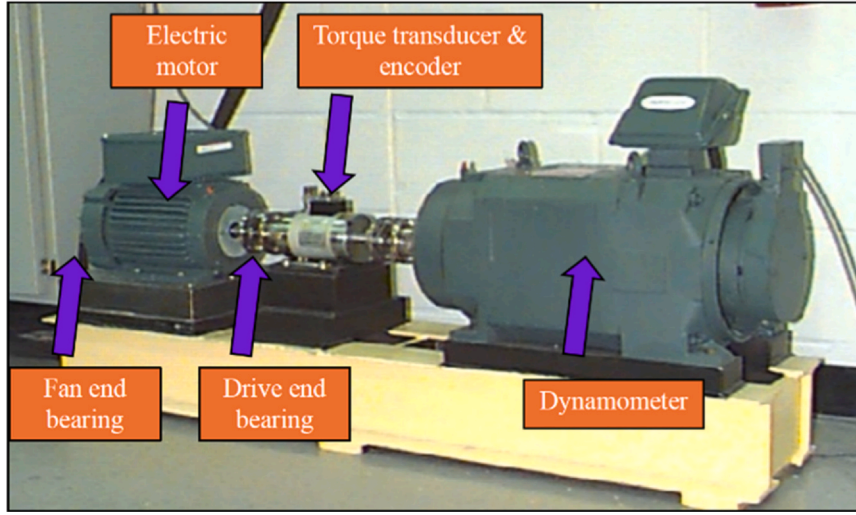


Fig. 5. The CWRU test platform.

Table 1
Experimental setup of the CWRU bearing dataset.

Class	Condition	Fault size (inch)	Training	Test	Length
C0	Normal	/	50	100	2048
C1	Inner ring wear	0.007	50	100	2048
C2	Outer ring wear	0.007	50	100	2048
C3	Rolling element wear	0.007	50	100	2048

Table 2
The main training parameters of the model.

Learning rate	Batch size	Epochs	η	λ
0.001	8	1000	0.1	1×10^{-5}

features of the original samples obtained. The calculations of the time-domain and frequency-domain characteristics of Z are shown in Table 3.

4. Results and discussion

4.1. Feature analysis

Based on the clearly defined sample characteristic dimension system in Fig. 6, the study first conducted quantitative calculations of physical characteristics for all experimental samples, ensuring that the characteristic values of each sample in each dimension were obtained through standardized quantification methods, laying a data foundation for subsequent correlation analysis. Based on this, the Pearson correlation coefficient was used as the metric to measure the degree of correlation between features, and sample feature correlation heatmaps and bubble charts were constructed. The results are shown in Fig. 6.

To accurately determine the strength of the correlation between features, the experiment set clear quantitative standards. The situations where the absolute value of the correlation coefficient is significantly high are defined as strong correlations, and based on this, feature combinations with a high degree of linear correlation are selected. Meanwhile, for cases where the correlation coefficient is negative, its physical meaning is that there is an inverse linear relationship between the corresponding variables.

Based on the experimental parameter design scheme determined in the experiment and the sample characteristic definition system defined in Fig. 6. In this experiment, the training data size was set at 50 samples per category. The focus was on comparing the training loss evolution process of the CN-LLM-KGN model under three different feature acquisition methods, in order to verify the impact of feature input methods on the stability of model training. The experimental results are shown in Fig. 7. The blue curve represents the total loss of the model after introducing L2 regularization, while the orange curve represents the marginal loss of TransE. These two curves jointly reflect the loss variation characteristics of the model at different training stages.

From the overall trend of the curves, it can be seen that the model training processes corresponding to the three feature acquisition schemes all exhibit excellent stability. On the one hand, both the total loss and the TransE marginal loss show a steady decline as the number of training rounds increases. There was no significant fluctuation or oscillation during the later stage of training, and no signs of overfitting, such as loss rebound, appeared. This indicates that regardless of the method used for feature acquisition, the output feature information can provide effective learning signals for the CN-LLM-KGN model, enabling the model to continuously optimize its parameters. On the other hand, the introduction of L2 regularization effectively suppresses the parameter oscillation problem during the model training process, preventing instability in training due to excessive parameter updates, and laying a crucial foundation for the performance reliability of the subsequent model inference stage.

4.2. Result analysis

To avoid the possible interference of randomness in a single reasoning result on the experimental conclusion, and to enhance the reliability and persuasiveness of the results. This study obtained the

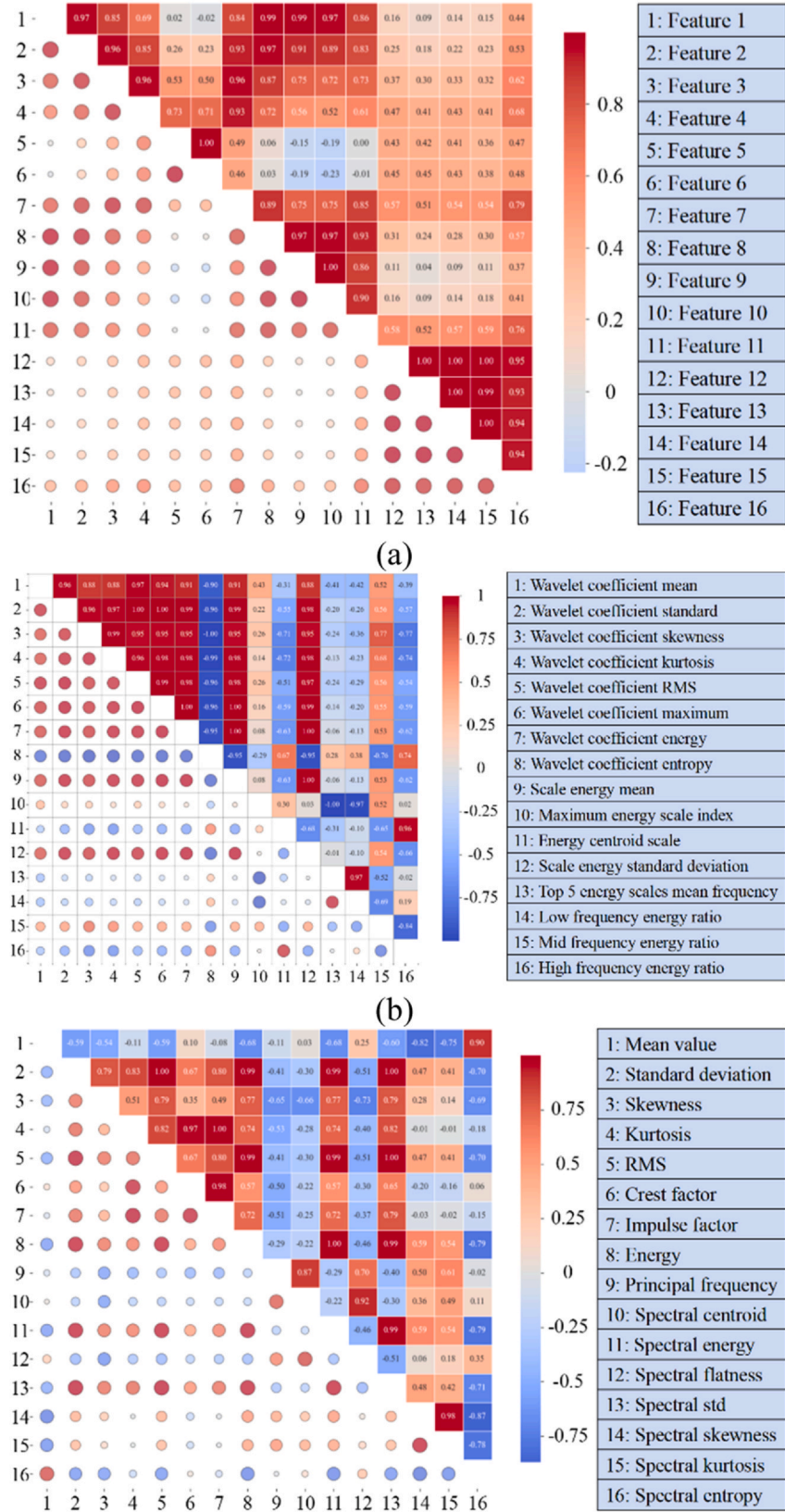


Fig. 6. The correlation heatmap and bubble diagram: (a) H , (b) T , (c) Z .

Table 3

The calculation for the time-domain and frequency-domain characteristics of Z.

Time-domain	Equation	Frequency-domain	Equation
Mean value	$\mu = \frac{1}{N} \sum_{i=1}^N x_i$	Principal frequency	/
Standard deviation	$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \mu)^2}$	Spectral centroid	$SC = \frac{\sum_{k=1}^M f_k \cdot P(f_k)}{\sum_{k=1}^M P(f_k)}$
Skewness	$S = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^3$	Spectral energy	$SE = \sum_{k=1}^M P(f_k)$
Kurtosis	$K = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^4 - 3$	Spectral flatness	$SF = \frac{\exp\left(\frac{1}{M} \sum_{k=1}^M \ln P(f_k)\right)}{\frac{1}{M} \sum_{k=1}^M P(f_k)}$
RMS	$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$	Spectral std	$S\sigma = \sqrt{\frac{\sum_{k=1}^M (f_k - SC)^2 \cdot P(f_k)}{\sum_{k=1}^M P(f_k)}}$
Crest factor	$CF = \frac{\max(x_i)}{RMS}$	Spectral skewness	$SS = \frac{\sum_{k=1}^M (f_k - SC)^3 \cdot P(f_k)}{(S\sigma)^3 \cdot \sum_{k=1}^M P(f_k)}$
Impulse factor	$IF = \frac{\max(x_i)}{\frac{1}{N} \sum_{i=1}^N x_i }$	Spectral kurtosis	$SK = \frac{\sum_{k=1}^M (f_k - SC)^4 \cdot P(f_k)}{(S\sigma)^4 \cdot \sum_{k=1}^M P(f_k)} - 3$
Energy	$E = \sum_{i=1}^N x_i^2$	Spectral entropy	$SE = -\sum_{k=1}^M p_k \cdot \ln p_k$

results through multiple independent reasoning experiments, and based on the statistical results, a confusion matrix, as shown in Fig. 8, was drawn. Among them, Fig. 8(a), (b), and (c) respectively correspond to the confusion matrix results of three different feature acquisition methods under the H , T , and Z feature systems.

From the visualization results of the confusion matrix, it can be directly observed that under the same experimental conditions, the feature acquisition method proposed in this study performed better in terms of recognition accuracy for most samples. The values on the main diagonal of its confusion matrix are significantly higher than those of other feature acquisition methods. The number of misclassified samples was significantly smaller, which fully demonstrated the superiority of the proposed method in the fault identification task.

Meanwhile, in order to further explore the performance details of the specific feature acquisition method, this study conducts an in-depth analysis of the feature acquisition method corresponding to Fig. 8(a). The box plot of the reasoning results under this feature scheme is shown in Fig. 9. It is used to investigate the differences in reasoning effects for samples of different categories.

From the distribution characteristics of the box plot in Fig. 9, it can be seen that this feature acquisition method performs exceptionally well in the reasoning process for C3-type samples. The core data representing the reasoning accuracy in the box plot are all at a relatively high level, and the data dispersion is small, indicating that the model has high accuracy and strong stability in reasoning for C3-type samples. In contrast, its reasoning effect on C1-type samples is significantly poorer. Not only is the core reasoning accuracy index significantly lower than that of the C3-type, but the data distribution is also scattered, with many outliers deviating from the normal range. This indicates that the model has weak reasoning ability and insufficient stability when dealing with C1-type samples.

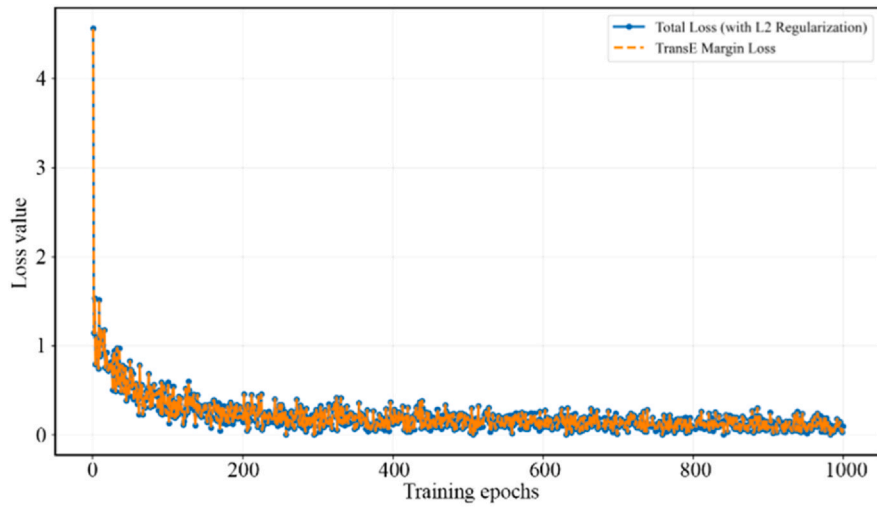
This conclusion can be directly confirmed from the confusion matrix in Fig. 8(a). On the main diagonal of the confusion matrix, the values corresponding to the C3-type samples account for an extremely high proportion, while the error values on the off-diagonal are very few. However, the proportion of correct classification values for the C1-type samples on the main diagonal is significantly lower. On the non-diagonal positions, there are more values indicating misclassification to other categories, which is highly consistent with the differences in reasoning effect revealed by the box plot.

To quantify the impact of different feature acquisition methods on the model performance, a comparative analysis was conducted using C3-type fault samples as the research object. The results are shown in Table 4. The experiment set up training data scales of 50 samples for each category and 30 samples for each category. From the two core dimensions of reasoning accuracy and reasoning efficiency, evaluate the performance of each feature scheme in the CN-LLM-KGN model.

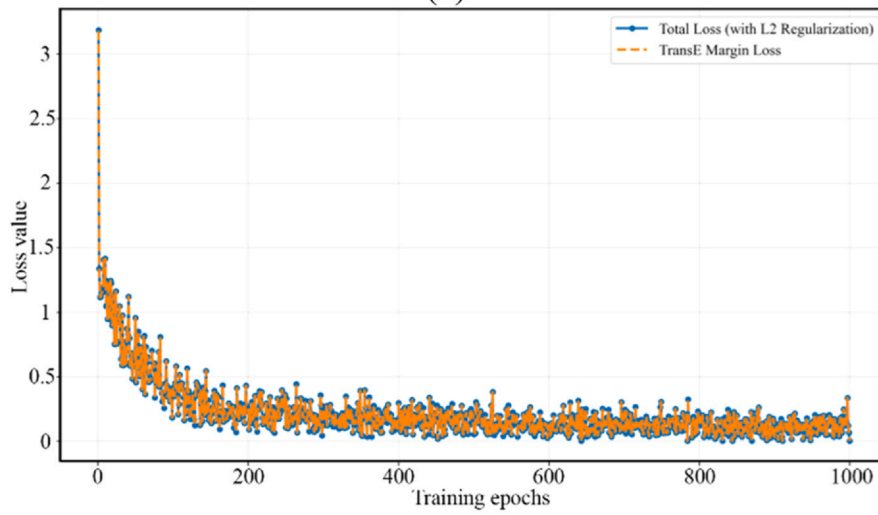
The results show that the features extracted using the CWT demonstrate significant advantages in both data scales. In each of the 50-sample training scenarios of each category, the inference accuracy of the model for C3-type samples reached 99.94 %, and the single inference time was only 37 ms. Even in the case of a small sample size of 30 for each category, the accuracy rate still remained at a high level of 99.39 %, and the inference time only slightly increased to 39 ms. This result fully demonstrates that the features obtained through continuous wavelet transformation can accurately capture the key physical information of C3-type faults, supporting the high robustness of the model in scenarios with few-shot samples.

During the knowledge graph enhancement stage, the triples completed by the large language model revealed the profound causal relationships among the fault parameters and constructed a clear fault propagation path. As shown in Fig. 10, taking the inner ring fault as an example, the causal path is mainly divided into three dimensions: speed, load, and amplitude. This figure visually presents the strength of the causal relationship between each operating parameter and the failure of the inner ring of the bearing. The figure highlights the key causal paths that are most similar to the fault. The 2 horsepower load is the primary factor causing inner ring failures, with an associated strength of 0.3395, which is significantly higher than that of other load conditions. This discovery is highly consistent with the input fault characteristic information, verifying the ability of the model to accurately identify the key causes of faults under the experimental conditions of this study, and providing interpretable evidence in support of the location of the root causes of faults.

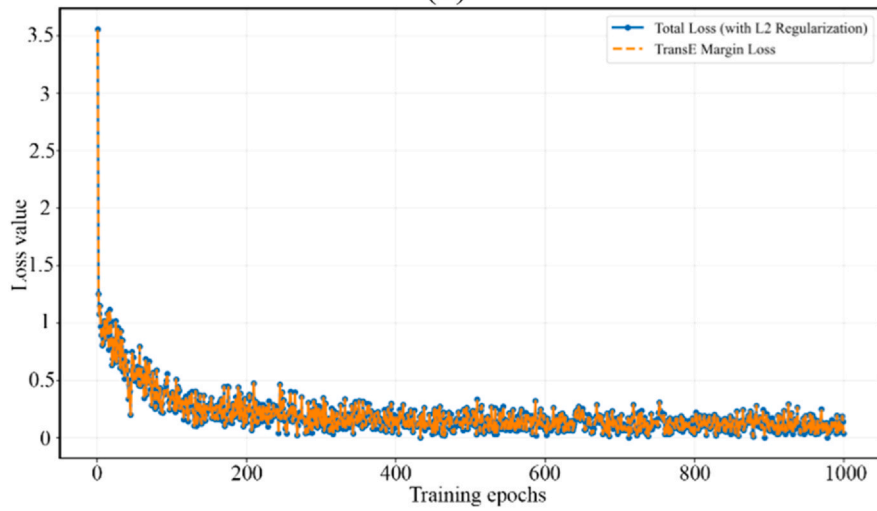
Finally, to systematically analyze the regulatory effect of the causal loss weight η on the inference results of the model. A specific experiment was designed to analyze the sensitivity of the hyperparameters to η , and the results were presented in a visual form in Fig. 11. From the analysis results of the hyperparameters in Fig. 11, it can be clearly observed that as the value of the causal loss weight η changes, the



(a)



(b)



(c)

Fig. 7. The loss curve of the CN-LLM-KGN model: (a) H , (b) T , (c) Z .

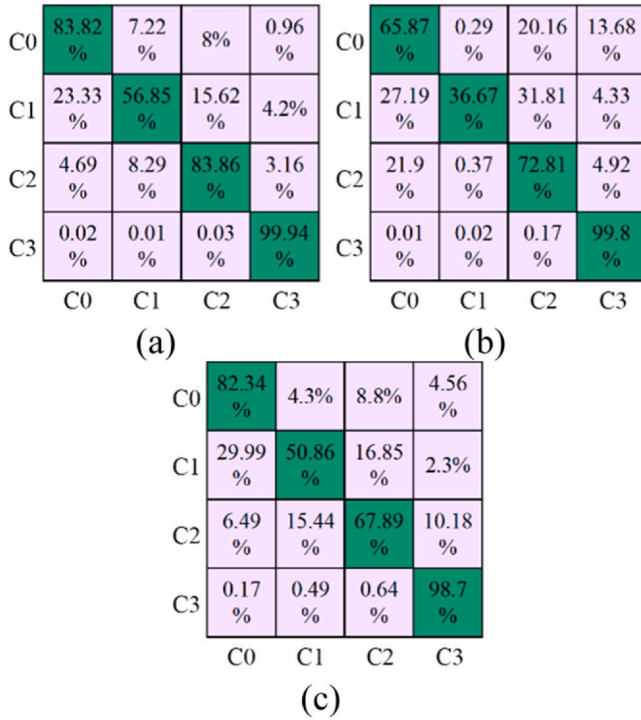
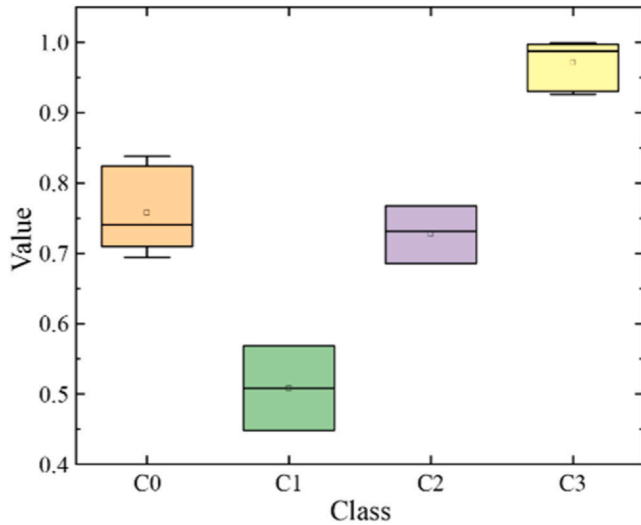
Fig. 8. The confusion matrix: (a) H , (b) T , (c) Z .

Fig. 9. The box plot of the proposed preprocessing method.

average inference performance of the model shows a significant fluctuating trend. Among them, when η takes the value of 0.1, the average inference result of the model reaches the optimal level, and the average inference accuracy is the highest, at 81.12 %. This indicates that this weight value can optimally balance the contribution of causal losses and other loss terms of the model, thereby maximizing the overall reasoning performance of the model.

4.3. Comparison and ablation experiment analysis

To rigorously evaluate the performance of the CN-LLM-KGN model proposed in this paper, it is compared with a series of representative advanced methods. These comparison benchmarks cover traditional machine learning, classic deep learning, the cutting-edge Transformer architecture, and the latest strong baseline model to ensure the comprehensiveness and fairness of the comparison.

Random Forest [33]: A classic ensemble learning algorithm. It enhances the generalization ability and robustness of the model by constructing multiple decision trees and aggregating their results.

Multilayer Perceptron (MLP) [34]: A fundamental feed forward neural network model. It can learn complex feature mappings through multiple layers of nonlinear transformation.

Attention long short-term memory (LSTM) [35]: This model combines the ability of LSTM networks to capture temporal dependencies with the capability of the attention mechanism to focus on critical information. This method is a strong baseline for processing sequential data like vibration signals.

Trustworthy lightweight multi-expert wavelet transformer (TLMW-former) [36]: An advanced temporal model based on the TLMW-former architecture. It utilizes the self-attention mechanism to globally capture long-range dependencies within signals, representing the latest paradigm for handling complex time series.

Ensemble multi-scale convolutional attention network (EMCAN) [37]: A recently proposed EMCAN that has demonstrated exceptional performance in fault diagnosis tasks. It was selected as a powerful, state-of-the-art direct competitor.

The comparison results are shown in Fig. 12. The proposed CN-LLM-KGN model in this paper achieved the best performance metric of 0.8112, significantly outperforming all the comparison methods. Specifically, its diagnostic accuracy rate has increased by more than 60 % compared to the random forest method, and by more than 220 % compared to the method using MLP. It is also significantly superior to the attentional LSTM, EMCAN, TLMW-former, and other advanced models proposed in recent years. This result fully demonstrates that the CN-LLM-KGN model has a significant advantage in accuracy for fault diagnosis tasks, verifying the effectiveness of its architecture design.

To quantitatively evaluate the contributions of each core component in the proposed CN-LLM-KGN model in this paper, and to verify its superiority over existing dedicated methods, a systematic ablation experiment and baseline comparison study were conducted in this section. The design and results of the ablation study are shown

Table 4
C3 type reasoning result.

Sample	Feature	C0	C1	C2	C3	Time
50	H	0.02 %	0 %	0.03 %	99.94 %	37 ms
	T	0.01 %	0.02 %	0.17 %	99.79 %	38 ms
	Z	0.17 %	0.449 %	0.63 %	98.7 %	38 ms
30	H	0.27 %	0.01 %	0.34 %	99.39 %	39 ms
	T	12.43 %	0.06 %	0 %	87.51 %	39 ms
	Z	17.6 %	2.68 %	1.17 %	78.55 %	39 ms

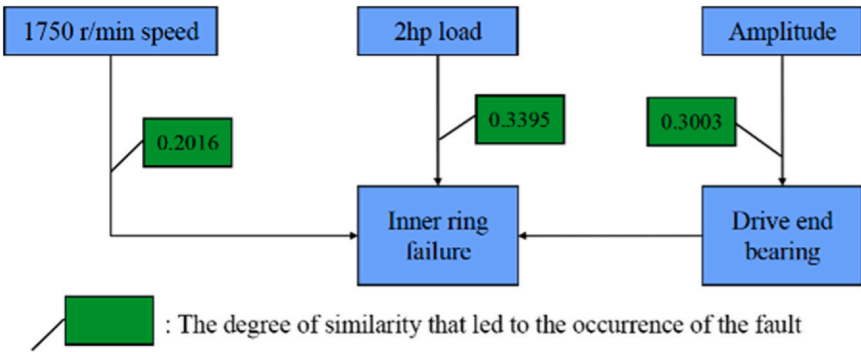


Fig. 10. Analysis of the path of inner ring failure.

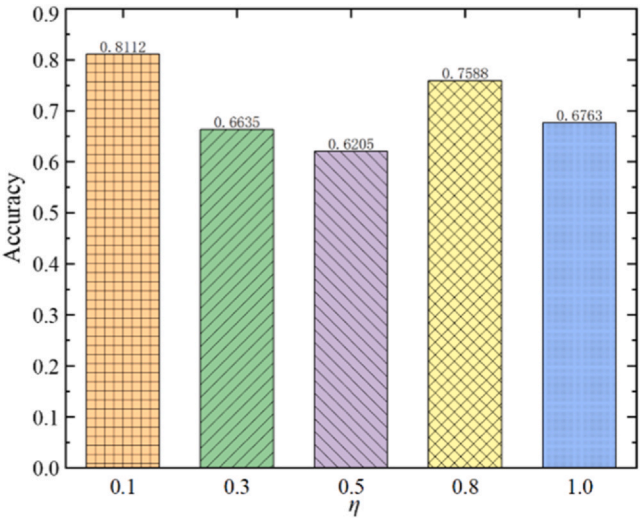


Fig. 11. Analysis of the hyperparameters for the weights of causal losses.

in Table 5. The ablation results clearly revealed the crucial roles of each component. Removing CN led to a sharp decline in performance to 57.8%, highlighting its fundamental role in extracting discriminative local features from the original vibration signal.

Table 5
The ablation design and reasoning results.

Methods	CN	L2	LLM	TransE	Reasoning result
Ablation 1	✓	✓	✓	✓	81.12 %
Ablation 2	×	✓	✓	✓	57.8 %
Ablation 3	✓	×	✓	✓	73.8 %
Ablation 4	✓	✓	×	✓	76.34 %
Ablation 5	✓	✓	✓	×	53.13 %

Disabling L2 regularization resulted in an accuracy drop to 73.8%, suggesting a decline in model robustness. This demonstrates its role in stabilizing the training process and mitigating overfitting to the training data. After excluding the knowledge module of the large language model, the accuracy rate was 76.34%, which confirmed the contribution of semantic prior knowledge to understanding complex fault chains. However, replacing the TransE reasoning module led to a significant drop in performance to 53.13%, proving the indispensability of structured knowledge graph reasoning in the fundamental cause localization process. The comprehensive results indicate that the outstanding performance of CN-LLM-KGN does not stem from a single component but is the result of the effective collaboration and deep integration of all core components.

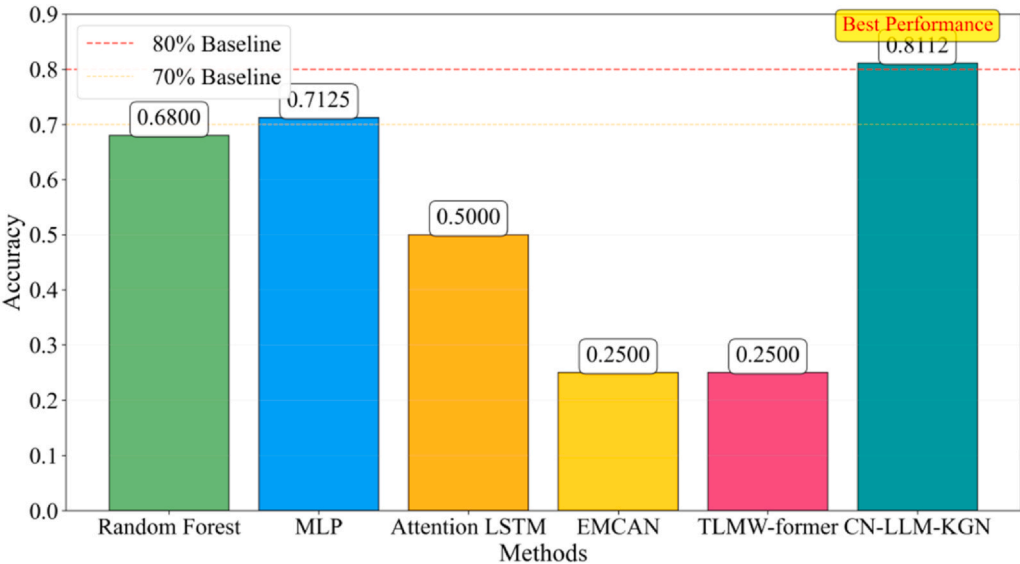


Fig. 12. Compared with other diagnostic methods.

5. Conclusions

This study addresses the issues of insufficient causal relationship mining and the limited reasoning ability of knowledge graphs in traditional mechanical fault diagnosis. A CN-LLM-KGN causal reasoning fault diagnosis method integrating CNs, LLMs, and KGNs is proposed. Through systematic feature optimization and comprehensive performance verification experiments, the following conclusions were drawn:

- (1) The time-frequency feature extraction method based on continuous wavelet transform can effectively capture the fault-sensitive information in mechanical monitoring signals, providing a highly discriminative and high-dimensional feature representation for causal discovery and graph construction.
- (2) By introducing L2 regularization and the marginal loss constraint based on TransE, the model can effectively learn the discriminative features and structured knowledge representations in the data, avoiding overfitting and improving the adaptability of diagnosis.
- (3) By integrating the data-driven causal mining mechanism and the knowledge-guided reasoning mechanism, the model achieves high-precision fault classification while also possessing good interpretability and causal traceability capabilities.
- (4) Finally, in the root cause localization task, an accuracy rate of 81.12% was achieved, providing a reliable and interpretable technical approach for intelligent diagnosis of rotating machinery systems.

Although the CN-LLM-KGN method proposed in this study demonstrated superior performance in fault diagnosis and root cause localization tasks, it still has some limitations.

- (1) This method has been verified to be effective on a single standard dataset. However, its generalization ability in complex and variable conditions as well as across different devices still needs further evaluation. In the future, a benchmark dataset across various operating conditions and devices will be constructed, and domain adaptation techniques will be explored to enhance the generalization performance of the model.
- (2) The current construction of knowledge graphs still relies to some extent on expert knowledge and manual definitions, and there are deficiencies in terms of automation and scalability. In the future, research will be conducted to utilize large language models to automatically extract and evolve knowledge from unstructured text, in order to enhance the automation level of knowledge graph construction.
- (3) Despite the adoption of efficient fine-tuning techniques, the deployment efficiency and real-time performance of large models remain a challenge in resource-constrained edge scenarios. In the future, we will explore technologies such as model compression and knowledge distillation to develop diagnostic-specific models that are more lightweight and better suited for deployment in industrial settings.

CRedit authorship contribution statement

Zhen Guo was responsible for model establishment, data analysis, and wrote the manuscript; Zhiping Liu, Edgar Estupiñan, and Diego Cabrera were in charge of the guidance and final check. All authors read and approved the final manuscript.

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Declaration of Competing Interest

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